RapidSmith: A Java Based CAD Tool For MODERN XILINX FPGAs

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The RapidSmith Project from Brigham Young University, USA, is a Java package that contains the tools and application program interfaces (APIs) for providing a unique platform to work out experimental ideas and algorithms on modern Xilinx field-programmable gate arrays (FPGAs). The tool allows users to import XDL/NCD, manipulate, place, route and export designs among a variety of design transformations.

As a tool based on Xilinx Design Language (XDL), RapidSmith provides a human-readable file format equivalent to Xilinx proprietary Netlist Circuit Description (NCD). Through xdl executable, RapidSmith can be used to modify designs at any point in ISE's design flow or even create one from scratch. Packages present in RapidSmith allow parsing and exporting of bitstreams at packet level, and representating frames and configuration blocks in the provided data structures. The latest release of the software, version 0.5.2, is bundled with this month's EFY Plus DVD.

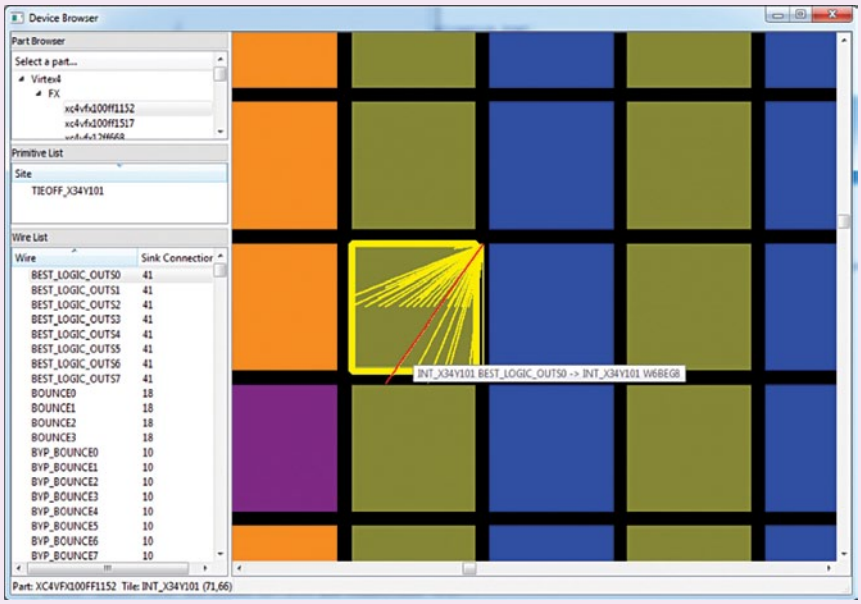


Fig. 1: The wires available in a tile and the connection name in the tooltip message

The motivation behind

Project administrators, Christopher Lavin and team, have described the motivation behind the development of RapidSmith in their document titled 'Rapid Prototyping Tools for FPGA Designs: RapidSmith.' They felt that the compilation time associated with the FPGA design was very high in comparison with what is required for software systems. A faster compilation process for FPGAs ensures improved productivity for designers as more designs and debugging could be completed within the stipulated time.

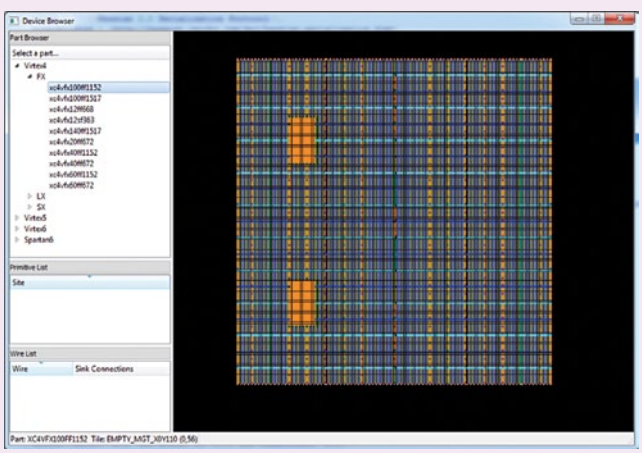


Fig. 2: Xilinx Virtex4 family supporting RapidSmith

Moreover, devices get bigger and bigger with every generation, thereby resulting in no significant improvement in FPGA implementation times.

Although, the target hardware is often available for design-verification during much of the development process, it often goes unused as there is no fast method for its proper utilisation for verification during the design process. Therefore RapidSmith has been developed as a rapid mapping mechanism for design onto an existing FPGA platform to perform rapid design and debug cycles. This, in turn, provides a more interactive development environment for FPGAs.